

Revised Load Interconnection System Impact Study Report

For Queue Project Q18-0930

Ralston 138 kV
March 19, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0930. The Interconnection Customer proposes a 26.6 MW large load (data center) facility interconnecting at the Ralston 138 kV substation in the Pennsylvania area. The request was received on January 24, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 381 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$50,685,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
HILLTOP - SULPHUR SPRINGS 138.0 kV	STRAWBERRY BELT - HILLTOP 345.0 kV	183.9	302	12.0
HILLTOP 345 kV/138 kV Autotransformer	HILLTOP - STRAWBERRY BELT 345.0 kV L	178.7	389	14.1
DAIRVILLE - SULPHUR SPRINGS 138.0 kV L	STRAWBERRY BELT - HILLTOP 345.0 kV	129.0	192	9.2
TALENENERGY MARTINS CREEK LLC WILLIAMSPORT SUBSTATION 345 kV/138 kV XFMR	TALENENERGY MARTINS CRK LLC WILLIAMSPORT - MONTOUR STATION 345.0 kV	114.1	422	39.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SULPHUR SPRINGS - HILLTOP 138.0 kV Line	Rebuild line with higher-capacity conductor	\$4,752,000	100% (\$4,752,000)
HILLTOP 345 kV/138 kV Autotransformer	Replace transformer with higher-rated unit	\$15,586,000	100% (\$15,586,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SULPHUR SPRINGS - DAIRYVILLE 138.0 kV	Rebuild line with higher-capacity conductor	\$13,320,000	100% (\$13,320,000)
TALENERGY MARTINS CREEK LLC WILLIAMSPORT SUBSTATION 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$17,027,000	100% (\$17,027,000)
	Total allocated to project		\$50,685,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.